

Exp No:4 QUERYING LEDGER DATA WITH HYPERLEDGER FABRIC**AIM**

To query and retrieve ledger data from a Hyperledger Fabric blockchain network using JavaScript chaincode, and verify asset information stored in the world state database.

REQUIREMENTS

- Kali Linux / Ubuntu
- Docker and Docker Compose
- Node.js (Version 18 or later) and npm
- Git
- Hyperledger Fabric Samples, Binaries, and Docker Images

PROCEDURE**1. Navigate to the Hyperledger Fabric test network directory:**

```
cd ~/fabric-dev/fabric-samples/test-network
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ cd ~/fabric-dev/fabric-samples/test-network
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ █
```

2. Start the Fabric network and create the application channel:

```
./network.sh up createChannel -ca
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -ca
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with crypto from 'Certificate Authorities'
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
CA_LOCAL_VERSION=v1.5.17
CA_DOCKER_IMAGE_VERSION=v1.5.17
Generating certificates using Fabric CA
[+] up 4/4
✓Network fabric_test Created 0.0s
✓Container ca_org2 Started 0.3s
✓Container ca_orderer Started 0.4s
✓Container ca_org1 Started 0.4s
+ fabric-ca-client getcainfo -u https://admin:adminpw@localhost:7054 --caname ca_org1 --tls.certfiles /home/naveen/fabric-dev/fabric-samples/test-network
```

3. Deploy the JavaScript Asset Transfer chaincode:

```
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-
basic/chaincode-javascript
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.s
h deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode
-javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-bas
ic/chaincode-javascript --lang node --label basic_1.0
+ res=0
Chaincode is packaged
```

4. Set the Orderer CA:

```
export
ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/orderer.e
xample.com/msp/tlscacerts/tlsca.example.com-cert.pem
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ export ORDE
RER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/ordere
r.example.com/msp/tlscacerts/tlsca.example.com-cert.pem
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ █
```

5. Initialize the ledger with sample asset data:

```
peer chaincode invoke -o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls --cafile $ORDERER_CA \
-C mychannel -n basic \
```

```
-c '{"function":"InitLedger","Args":[]}'
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ export ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls --cafile $ORDERER_CA \
-C mychannel -n basic \
-c '{"function":"InitLedger","Args":[]}'
2026-09-01 17:01:43.819 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery
-> Chaincode invoke successful. result: status:200
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

6. Query all asset records stored in the ledger:

```
peer chaincode query -C mychannel -n basic \
-c '{"Args":["GetAllAssets"]}'
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset1"]}'
{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

7. Query the specific asset asset1:

```
peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset1"]}'
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset1"]}'
{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

8. Create a new asset asset20:

```
peer chaincode invoke -o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls --cafile $ORDERER_CA \
-C mychannel -n basic \
-c '{"function":"CreateAsset","Args":["asset20","Green","8","Arun","1500"]}'
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls --cafile $ORDERER_CA \
-C mychannel -n basic \
-c '{"function":"CreateAsset","Args":["asset20","Green","8","Arun","1500"]}'
--peerAddresses localhost:7051 \
--tlsRootCertFiles "${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles "${PWD}/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
--waitForEvent
2026-09-01 17:14:06.232 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [0aef9f170983bf81cb4c36bbe55963e5c02c24d1da73eea341d325ae8fe0ff9b] committed with status (VALID) at localhost:7051
2026-09-01 17:14:06.232 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [0aef9f170983bf81cb4c36bbe55963e5c02c24d1da73eea341d325ae8fe0ff9b] committed with status (VALID) at localhost:9051
2026-09-01 17:14:06.232 UTC 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:{"ID":"asset20","Color":"Green","Size":8,"Owner":"Arun","AppraisedValue":1500}
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

9. Query the newly created asset asset20:

```
peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset20"]}'
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset20"]}'
{"AppraisedValue":1500,"Color":"Green","ID":"asset20","Owner":"Arun","Size":8}
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

10. Query a non-existing asset asset50 to verify error handling:

```
peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset50"]}'
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic \
-c '{"Args":["ReadAsset","asset50"]}'
Error: endorsement failure during query. response: status:500 message:"The asset asset50 does not exist"
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$
```

11. Stop the Fabric network:

```
./network.sh down
```

```
naveen@LAPTOP-258I2DTL:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] down 10/10
✓Container orderer.example.com      Removed      0.6s
✓Container ca_org1                  Removed      3.6s
✓Container ca_orderer               Removed      3.7s
✓Container ca_org2                  Removed      3.4s
✓Container peer0.org1.example.com    Removed      1.0s
✓Container peer0.org2.example.com    Removed      1.1s
✓Volume compose_peer0.org2.example.com Removed      0.0s
✓Volume compose_peer0.org1.example.com Removed      0.0s
✓Network fabric_test                Removed      0.2s
✓Volume compose_orderer.example.com  Removed      0.0s
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
```

EXPECTED OUTPUTS

Ledger initialization: Ledger Initialized Successfully

asset1 query: ID: asset1 | Color: blue | Size: 5 | Owner: Tomoko | AppraisedValue: 300

asset20 creation: Asset asset20 created successfully

asset20 query: ID: asset20 | Color: Green | Size: 8 | Owner: Arun | AppraisedValue: 1500

Non-existing asset: Error: The asset asset50 does not exist.

Network shutdown: Network stopped successfully

RESULT

The ledger data stored in the Hyperledger Fabric blockchain network was successfully queried using JavaScript chaincode. Asset records were retrieved from the world state database, specific asset details were verified, and the newly created asset was successfully queried, demonstrating efficient data retrieval and verification in a permissioned blockchain environment.